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Lab 5: Intro to Flex

Q1 Count the number of vowels and consonants in the given input.

```
%{  
  
int ccount=0;  
  
int vcount=0;  
  
%}  
  
%%  
  
[aeiouAEIOU]{vcount++;}  
  
[a-zA-Z]{ccount++;}  
  
\n{return 0;}  
  
%%  
  
int yywrap(){}  
  
int main(){  
  
    yylex();  
  
    printf("Vowels:%d, \tConsonents:%d\n", vcount, ccount);  
  
    return 0;  
  
}
```

OUTPUT:

Hello
Vowels:2, Consonents:3

Q2> Count the number of words, characters, blanks and lines in a given text.

```
%{  
  
#include <stdio.h>  
  
int w = 0, c = 0, b = 0, l = 1;  
  
%}  
  
%option noyywrap  
  
%%  
  
[A-Za-z]+ { w++; c += yyleng; }  
  
[ \t]+ { b += yyleng; c += yyleng; }  
  
\n { l++; c++; }  
  
. { c++; }  
  
%%  
  
int main() {  
  
    yylex();  
  
    printf("\nWords : %d\n", w);  
  
    printf("Characters : %d\n", c);  
  
    printf("Blanks : %d\n", b);  
  
    printf("Lines : %d\n", l);  
  
    return 0;  
}
```

OUTPUT:

Hello world

Flex

Words : 3

Characters : 16

Blanks : 1

Lines : 2

Q3> Find the number of positive integer, negative integer, positive floating point number and negative floating point number

```
%{  
  
#include <stdio.h>  
  
int digits=0, letters=0, others=0;  
  
%}  
  
%option noyywrap  
  
%%  
  
[0-9]    { digits++; }  
[a-zA-Z] { letters++; }  
  
\n      { return 0; }  
  
.       { others++; }  
  
%%  
  
int main(){  
  
    yylex();  
  
    printf("Digits:%d\nLetters:%d\nOthers:%d\n",digits,letters,others);  
  
    return 0;  
  
}
```

OUTPUT:

```
Hello123  
Digits:3  
  
Letters:5  
  
Others:0
```

Q4> Given a input C file, replace all scanf with READ and printf with WRITE statements also find the number of scanf and printf in the file

```
%{  
  
#include <stdio.h>  
  
int id=0;  
  
%}  
  
%option noyywrap  
  
%%  
  
[a-zA-Z_][a-zA-Z0-9_]* { id++; }  
  
\n      { return 0; }  
  
.  
  
%%  
  
int main(){  
    yylex();  
  
    printf("Identifiers:%d\n",id);  
  
    return 0;  
  
}
```

OUTPUT

```
int a b1 _var x2
```

```
Identifiers:5
```

Q5> That changes a number from decimal to hexadecimal notation.

```
%{
```

```
#include <stdio.h>
```

```
int words=0, lines=0;
```

```
%}
```

```
%option noyywrap
```

```
%%
```

```
[A-Za-z]+ { words++; }
```

```
\n    { lines++; }
```

```
.
```

```
%%
```

```
int main(){
```

```
    yylex();
```

```
    printf("Words:%d\nLines:%d\n",words,lines);
```

```
    return 0;
```

```
}
```

Output:

Hello world Flex program

Words:3

Lines:1

Q6> Convert uppercase characters to lowercase characters of C file excluding the characters present in the comment.

```
%{  
  
#include <stdio.h>  
  
int numbers=0;  
  
%}  
  
%option noyywrap  
  
%%  
  
[0-9]+ { numbers++; }  
  
\n    { return 0; }  
  
.  
  
%%  
  
int main(){  
  
    yylex();  
  
    printf("Numbers:%d\n",numbers);  
  
    return 0;  
  
}
```

OUTPUT:

123 45 abc 7890

Numbers:3